

Supplementary Material

Cooperation by Default: LLM Agents in the Collective-Risk Dilemma

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1 Models

Every game was played through one model-serving interface, with the same settings for all five models: temperature 0.7 unless a condition states otherwise, and an output limit of 3,000 tokens, raised automatically for a reply that reached the limit before its final **CONTRIBUTION**: line.

Name in the paper	Model	Provider	Identifier
Claude Haiku	Claude Haiku 4.5	Anthropic	claude-haiku-4-5-20251001
Gemini Flash-Lite	Gemini 3.5 Flash-Lite	Google	gemini-3.5-flash-lite
GPT Luna	GPT-5.6 Luna	OpenAI	gpt-5.6-luna
Qwen	Qwen3-235B-A22B-Instruct	Alibaba	qwen3-235b-a22b-instruct-2507
Grok	Grok 4.20, non-reasoning	xAI	grok-4.20-0309-non-reasoning

2 Prompts

Each round, every seat receives the prompt below, rendered for its own position, the current round and the full history, and answers with one contribution. No seat is given a persona and the running pool is not printed. The texts are generated from the program that played the games, not retyped.

2.1 Base prompt

The prompt one seat receives in round 3 at $p = 0.9$ (the history is an example). The two equal-split phrases, which the no-hint prompt deletes, are in bold.

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You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social
dilemma.
At the start, each player privately receives an endowment of 40 monetary units. Your final cash
payoff depends on the rules below. The game lasts exactly 10 rounds.
Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate
account". The only amounts allowed are 0, 2, 4.
The group's target: by the end of the game, the combined contributions of all 6 players must reach
at least 120 (an average of 2 per player per round).
- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for
example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% the group is
hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the
remaining 10% probability, each player keeps their remaining money.
- Money put into the climate account is gone for good and is never refunded, whether or not the
target is reached.
After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but
each player always appears in the same fixed position. Players cannot talk to or message each other.
Current state:
- Round 3 of 10.
- Your remaining money: 36.
Previous rounds (each player keeps the same position; you are position 3):
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Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.
Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.
Now decide your contribution for round 3. Allowed choices: 0, 2, 4.
Output only your decision as a final line:
CONTRIBUTION: <one of 0, 2, 4>

2.2 No hint

Identical to the base prompt except for the two lines below.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).

Changed to: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120.

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute.

2.3 Neutral wording

Identical to the base prompt except for the lines below. The normative or naming phrases are replaced; the equal-split phrases are kept. The pool line that the base prompt can show is also reworded, but no condition in the paper shows it.

Base: You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social dilemma.

Changed to: You are Player_3, player 3 of 6, in a group experiment.

Base: Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate account". The only amounts allowed are 0, 2, 4.

Changed to: Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

Changed to: The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

Base: - If the group does not reach 120: the computer runs a lottery: with probability 90% the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.

- Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.

Changed to: - If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.

- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

2.4 Neutral wording, goal stated

The neutral-wording prompt with one opening paragraph added. Full text:

You are a participant in a paid economics experiment. The only thing that matters to you is your own final cash payoff. There are no right or wrong answers; decide however you judge best serves your own outcome.

You are Player_3, player 3 of 6, in a group experiment.

At the start, each player privately receives an endowment of 40 monetary units. Your final cash payoff depends on the rules below. The game lasts exactly 10 rounds.

Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.

Current state:

- Round 3 of 10.
- Your remaining money: 36.

Previous rounds (each player keeps the same position; you are position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

2.5 Two rewordings

Both keep every rule, the equal-split phrases and the final instruction, and change only the phrasing.

Rewording 1.

In this study you take the role of Player_3, seated in position 3 among 6 participants, in a task that models a collective-risk social dilemma.

Every participant starts with a private endowment of 40 monetary units, and what you finally take home is set by the rules that follow. There are exactly 10 rounds, no more and no fewer.

In every round the 6 of you choose, at the same time and without seeing each other's choices, an amount to move into a common "climate account". The permitted amounts are 0, 2, 4.

What the group has to achieve: the contributions of all 6 participants, added up across the whole game, must come to 120 or more (**an average of 2 per player per round**).

- Should the group get to 120: each participant walks away, in cash, with whatever they held back (**for example, contributing 2 every round leaves you 20 at the end**).
- Should the group fall short of 120: a lottery is drawn. With probability 90% a disaster strikes and every participant loses all the money they still hold (final cash payoff 0); with the remaining 10% probability, everyone keeps what they still hold.
- Whatever enters the climate account stays there permanently and is refunded under no circumstances, target reached or not.

Once a round closes, the 6 amounts contributed in it are shown to everyone. Identities stay hidden, but each participant occupies one fixed seat throughout. Participants cannot talk to or message one another.

Where things stand:

- This is round 3 out of 10.
- Money still in your hands: 36.

Earlier rounds (seats never change; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:
CONTRIBUTION: <one of 0, 2, 4>

Rewording 2.

You have been assigned the role of Player_3 and hold position 3 out of 6 in a group task modelling a collective-risk social dilemma.

Each participant is granted, in private, a starting sum of 40 monetary units. The rules set out below determine how much of that sum you ultimately receive in cash. Play continues for precisely 10 rounds.

Round by round, the 6 participants independently and secretly nominate a sum to transfer into a jointly held "climate account". The admissible sums are 0, 2, 4, and no others.

The collective requirement: when play ends, the total transferred by the 6 participants must amount to no less than 120 (**an average of 2 per player per round**).

- In the event that the total reaches 120: every participant retains as cash the portion of their endowment that was never transferred (**for example, contributing 2 every round leaves you 20 at the end**).

- In the event that the total falls below 120: the computer conducts a draw. With probability 90% a disaster occurs and the residual holdings of every participant are forfeited in full (final cash payoff 0); with the complementary probability of 10%, each participant retains their residual holdings.

- Sums transferred into the climate account are irrecoverable and are never returned, irrespective of whether the requirement is met.

At the close of each round, the 6 amounts transferred during that round are disclosed to all participants. Names are withheld; however, every participant remains at the same position for the duration. Communication of any kind between participants is not permitted.

Present position:

- Round 3 of 10 is underway.
- Funds you still retain: 36.

Record of previous rounds (positions are fixed; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>